



JOATHAM JACOB

Energy & Decarbonization Engineer

Industrial energy efficiency · Marginal abatement cost modelling

· Energy management systems (ISO 50001)

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Indian national · Work-authorized in Italy · Open to relocation;

PROFILE

Energy and environmental engineer (B.Tech) with an M.Sc. in Environmental Change and Global Sustainability (100/110, full scholarship), specialised in identifying, costing and prioritising industrial energy-efficiency and decarbonization measures. Combines site-level energy engineering — auditing, ISO 50001 baselining and monitoring, waste-heat recovery, heat electrification, motors and drives — with techno-economic modelling (NPV, IRR, LCOE, marginal abatement cost) and regulatory fluency across the EU ETS, CBAM and CSRD/ESRS. 14 months of hands-on Scope 1–3 measurement for industrial and corporate clients across two Milan consultancies, now applied to the reduction side: which measures physically abate, at what cost, and under which conditions they become financially viable. Builds custom analytical tools in Excel, Python and React.

TECHNICAL PROJECTS

Independent, self-directed analyses built from public data — baselines reconstructed from verified regulatory sources, costs from engineering estimates. Models and methodology available on request.

Staged decarbonization roadmap — Acciaieria Arvedi, Cremona (Italy) (2026)

Reconstructed the physical Scope 1 baseline from EU ETS verified emissions — 434,844 tCO₂ (CY2023) — after resolving installation boundaries between the operator's Cremona and Trieste sites, then decomposed it across process-carbon, combustion and electrode sources. Showed the plant's third-party-certified net-zero position to rest on three accounting instruments and one physical reduction. Modelled a costed 2030 marginal abatement cost curve: ~190 ktCO₂/yr technically available at ~€11 per tonne of steel, under 2% of product value — with 59% of it concentrated in a single biocarbon measure requiring 36 kt/yr of certified biochar against a European market of tens of kilotonnes. Quantified ETS and CBAM liability rising from €6.9M to €32.6M/yr under free-allocation phase-out, against a net-short position of 82,536 allowances.

Marginal abatement cost analysis — mid-market EAF steel plant (2026)

Modelled eight interventions for a 500,000 t/yr plant against a 150,000 tCO₂e baseline at an 8% discount rate; identified negative-cost measures abating ~6,000 tCO₂e/yr for €2.7M, with sensitivity analysis on EU allowance prices locating the thresholds at which capital-intensive measures turn viable.

Interactive abatement cost and energy finance application (2026)

React application computing payback, NPV, IRR, LCOE and levelised cost of abatement dynamically from user inputs, with a custom waterfall chart and carbon price reference line. Built end to end with LLM-assisted development, deployed via GitHub Actions.

Energy performance monitoring and forecasting pipeline (2026)

Python pipeline ingesting hourly consumption data, computing a weather-normalised baseline through degree-day regression, detecting deviations from expected performance, and reporting through Power BI. Structured for repeatable auditable reporting under ISO 50001.

Field energy audits — 8 railway stations, Bihar (India) (2026)

25-day multi-site campaign: site surveys, load analysis, energy and water consumption measurement, and site-specific reporting with prioritised interventions.

PROFESSIONAL EXPERIENCE

ESG Supply Chain & Decarbonisation Intern — Industrious Global Technologies, Milan (Feb 2025 – Aug 2025)

Calculated company Scope 1 and 2 energy-related emissions for 2024 operations from primary consumption data; collected, validated and consolidated energy and operational data under formal quality controls. Assessed emission-reduction pathways, identified efficiency opportunities across operations, and maintained KPI tracking and reporting for management.

ESG Measurement Intern — Omtra S.r.l Benefit Corporation, Milan (Nov 2023 – May 2024)

Collected primary energy and process data on site at client facilities; participated in internal audits and third-party verification, testing reported data against source evidence. Benchmarked client performance against sector standards and supported client-facing consultation on efficiency and improvement measures.

ESG Analyst (Remote) — Earth Academy Global, Amstelveen, Netherlands (Sep 2021 – Nov 2022)

Analysis and internal review across corporate reporting frameworks for international clients. Developed and delivered technical training content on environmental solutions and renewable energy technologies.

EDUCATION

M.Sc. Environmental Change and Global Sustainability — Università degli Studi di Milano (2022–2025)

- Full academic scholarship. Final grade 100/110. Taught entirely in English.
- **Thesis:** Emission Reduction Commitments in Global Agriculture: A Comparative Analysis of Corporate Climate Disclosures.

B.Tech. Energy and Environmental Engineering — Tamil Nadu Agricultural University (2016–2021)

- **Coursework:** Energy auditing, Renewable energy systems, Thermodynamics, Environmental impact assessment.
- **Field experience:** Energy, water, and safety audits of 8 railway stations in Bihar, India; biogas plant performance evaluation and energy audit, Palakkad; solar power plant design and cost estimation.
- **Thesis:** Design and Fabrication of a Solar Powered Trash Compactor (published, TNAU 2020).

KEY SKILLS

- **Energy & decarbonization** — energy auditing, ISO 50001 (baselines, EnPIs, monitoring), MACC development, industrial energy systems (motors and VFDs, compressed air, steam, process heat), heat electrification, waste heat recovery, M&V principles
- **Analysis & finance** — techno-economic modelling (NPV, IRR, LCOE, levelised cost of abatement, payback, capital recovery), scenario and sensitivity analysis, regression-based baselining
- **Carbon & regulation** — GHG Protocol (Scope 1–3), ISO 14064-1/-2, project-level quantification, carbon pricing and allowance markets, CBAM, CSRD/ESRS, SBTi
- **Tools** — Excel (advanced), Power BI, Python (pandas), React, Tableau, PV Syst, AutoCAD, QGIS
- **Languages** — English (C1), Italian (B1)

CERTIFICATIONS

- GHG 202: Basics of Project-Level GHG Accounting — Greenhouse Gas Management Institute (in progress, expected August 2026), GHG Accounting: Corporate Standard — GHG Protocol (2026)
- ISO 50001 Energy Management Systems, Prepared to qualify as Lead Auditor within six months,
- Energy Audits, Energy Efficiency Fundamentals — Schneider Electric University (2026)
- IGBC Accredited Professional